

Nextflow Introduction

**Workflow
managers and why
to use them**

Rule one of
bioinformatics

Be lazy

What are workflow managers

Be lazy: use a workflow manager!

Workflow managers provide a framework for the creation, execution, and monitoring of pipeline. <...>

They simplify pipeline development, optimize resource usage, handle software installation and versions, and run on different compute platforms, enabling workflow portability and sharing.

Wratten et al. (2021) *Nature Methods*

Summary

Pipelines using workflow managers benefit users by being

- More **portable**
 - Can be used on a very wide range of infrastructure
- More **efficient**
 - Less for users to install
 - More optimising resource usage
 - Extremely parallelised

More
reproducible

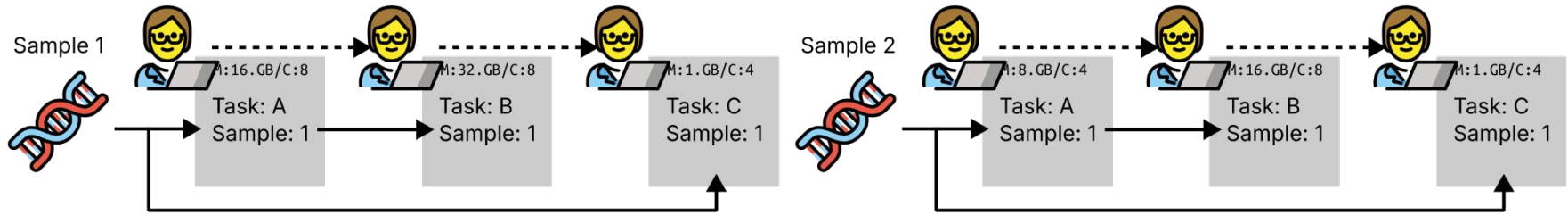
Than manual analyses, or 'custom' pipeline frameworks

Why are workflow managers
more efficient?

New Ph.D. student example

1. Writes a command
2. Puts it a `bash` script
3. Writes `sbatch` script that executes bash script
4. Submits the `sbatch` script
5. Repeatedly runs `squeue` to monitor job until completed
6. Writes next command
7. Puts it a new `bash` script
8. Writes new `sbatch` script that executes new bash script
9. Submits the new `sbatch` script
10. Repeatedly runs `squeue` to monitor job until completed
11. Writes next command
12. Puts it a new `bash` script
13. Writes new `sbatch` script that executes new bash script
14. Submits the new `sbatch` script
15. Repeatedly runs `squeue` to monitor job until completed

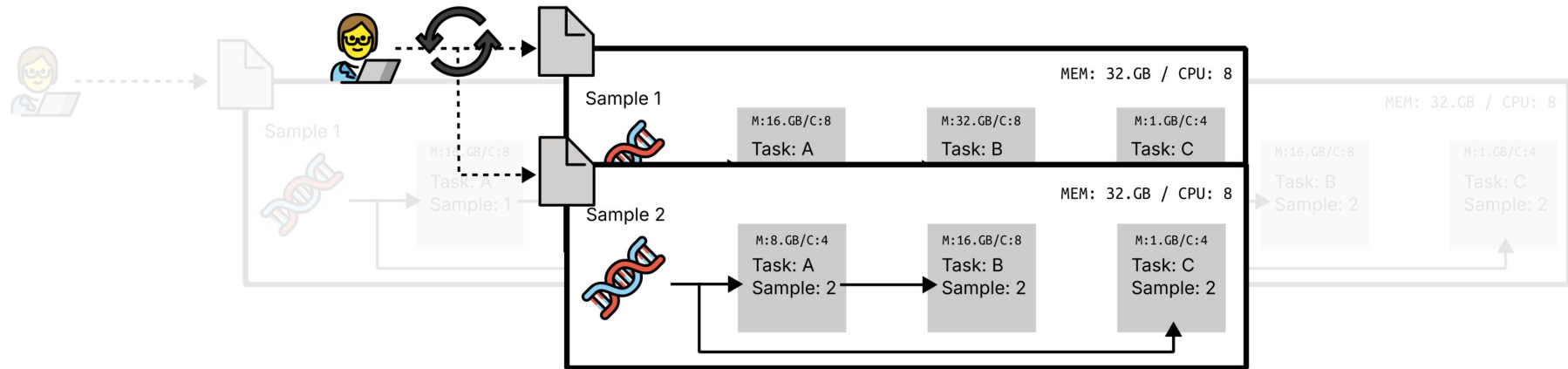
New Ph.D. student example (Graphical)



Second year Ph.D. example

1. Write a dynamic `bash` script
 - a. Supply input on the command line
 - b. Runs all commands of the analysis
 2. In the `sbatch` script, `for` loop to give all samples to the bash script
 3. Submit `sbatch` script
 4. Monitor until complete
- ✗ some of the jobs in the loop needed more memory than specified!

Second year Ph.D. example (Graphical)

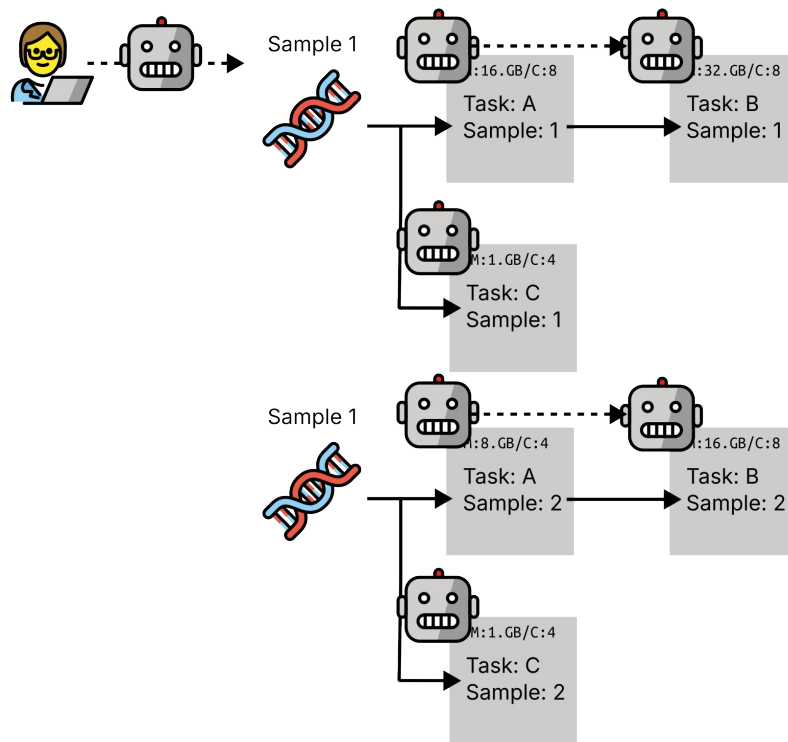


Ph.D. students in the know

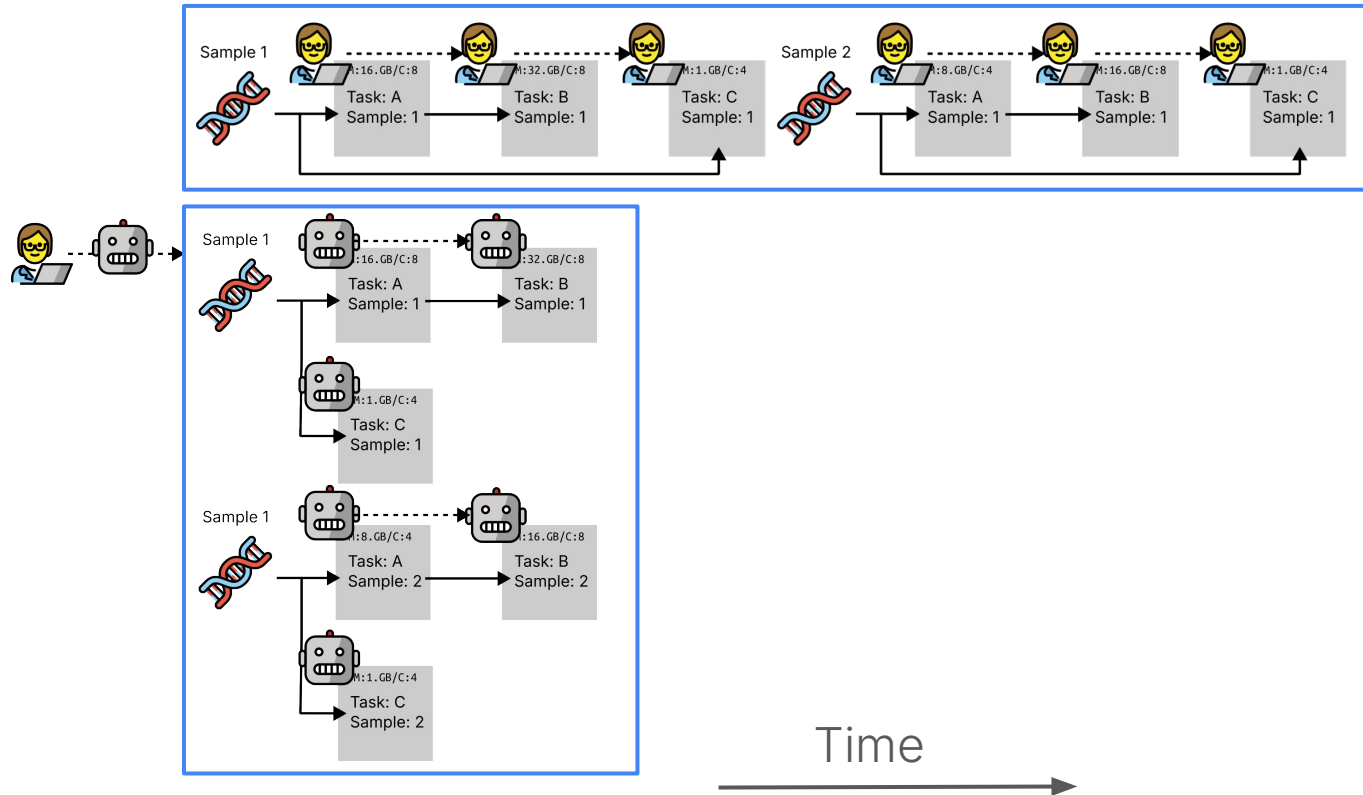
Use a workflow manager!

1. Execute a single command with the input and parameters
2. The workflow manager
 - a. Writes `bash` scripts for you
 - b. Executes jobs in optimal order for you
 - c. Handles file transfer between scripts for you
 - d. Handles writing of `sbatch` scripts for you
 - e. Handles re-execution of failed jobs for you
3. You wait for an email or notification saying success 🎉

Ph.D. students in the know (graphical)



Why are workflow managers more efficient?



Using computers to collect, store, analyze, and disseminate data and information



Large files

> 100 GB for one raw human genome...



Many languages

Bash, Python, R, PERL...



Complex interactions

Networks of software and their dependencies...

Reproducibility

Hidden reproducibility issues are like an iceberg



First, we tried to re-run the analysis with the code and data provided by the authors.

Second, we reimplemented the whole method in a Python package...



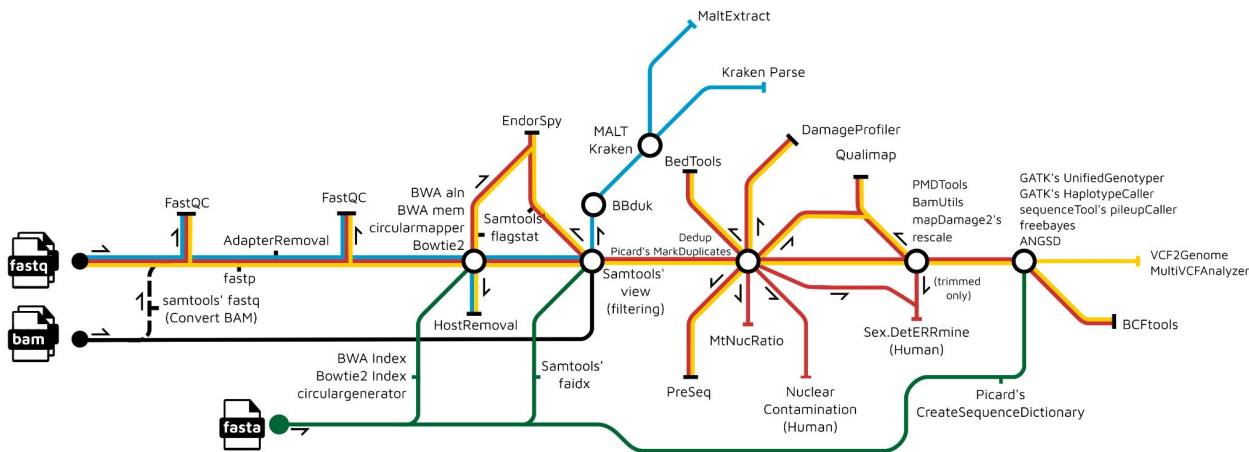
Experimenting with reproducibility:
a case study of robustness in bioinformatics
Kim et al., GigaScience (2018).

<https://doi.org/10.1093/gigascience/qiy077>



Nextflow

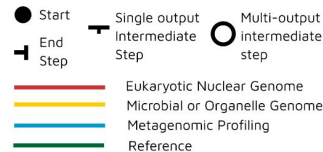
Managing modern workflows is complicated



nf-core/eager v2.4

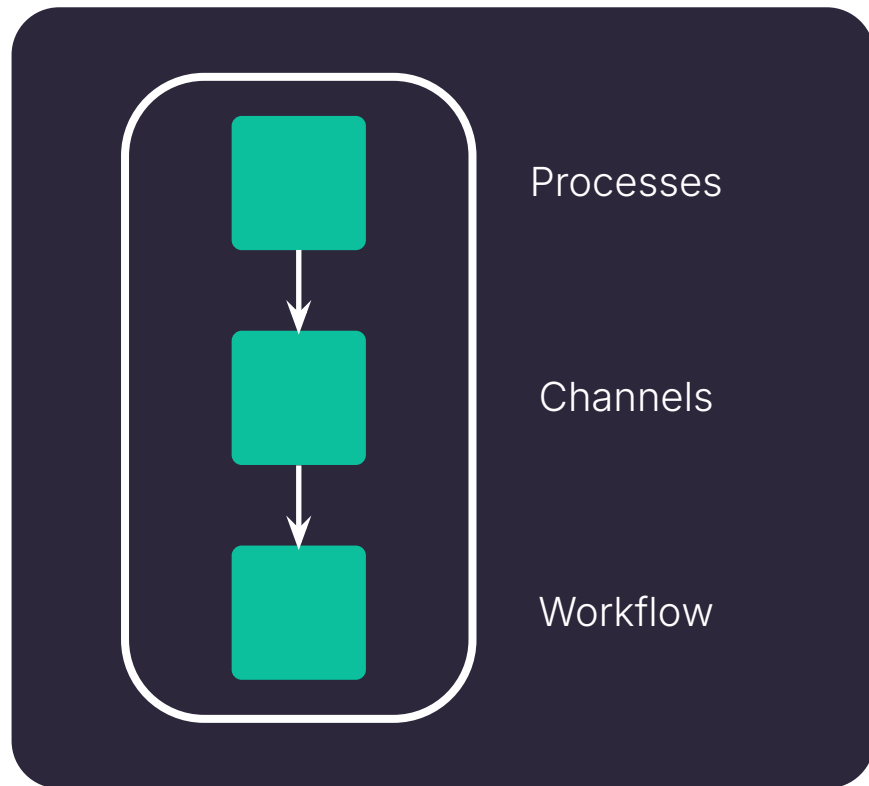
Example analysis pathways

Legend



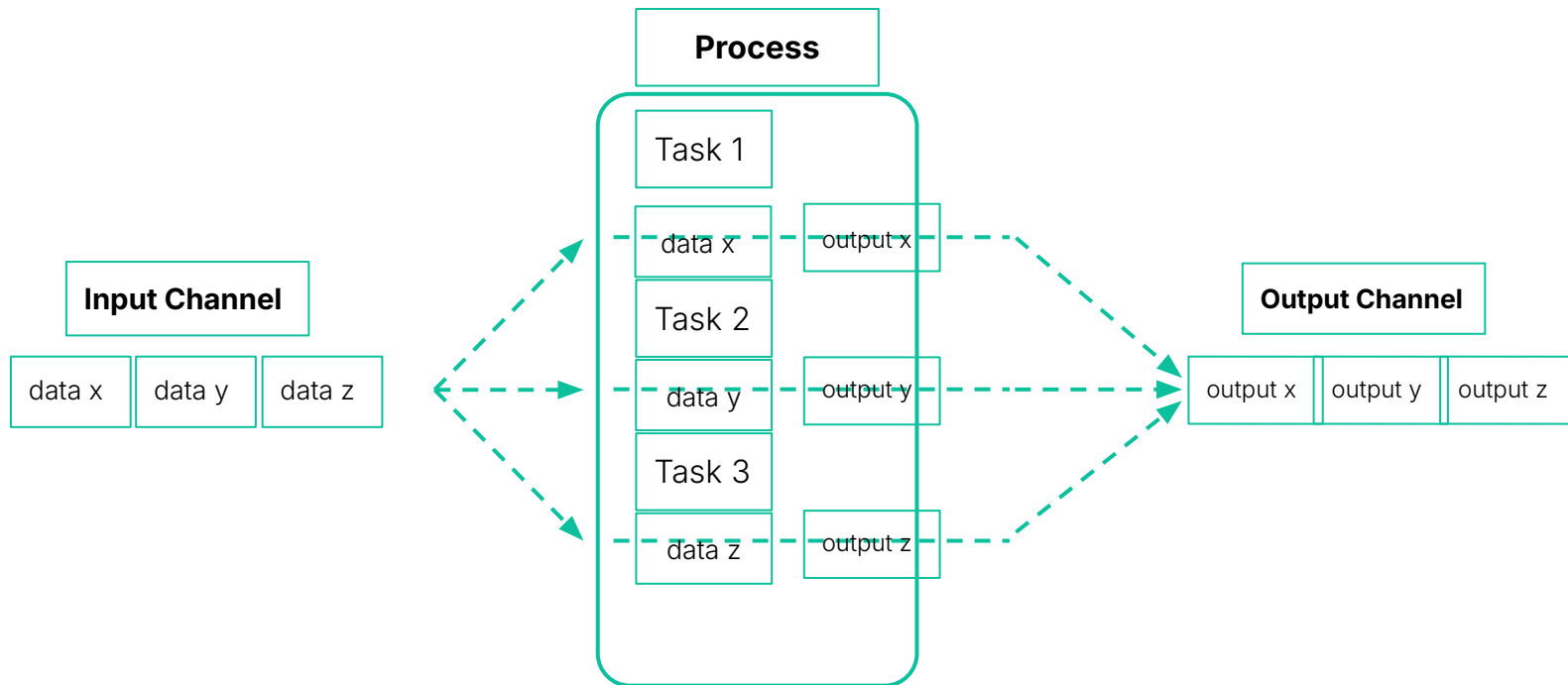
Nextflow

A reactive workflow framework and a programming DSL



Nextflow

A reactive workflow framework and a programming DSL



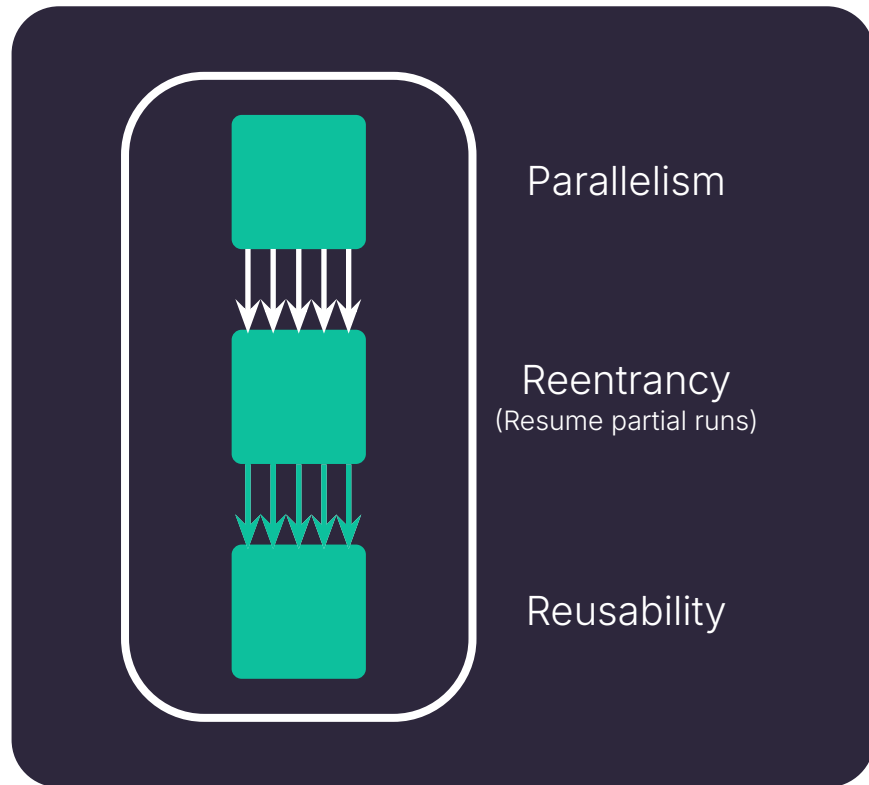
Nextflow

A reactive workflow framework and a programming DSL

```
process fastqc {  
  input:  
  path input  
  
  output:  
  path "*_fastqc.{zip,html}"  
  
  script:  
  ""  
  fastqc -q $input  
  ""  
}  
  
workflow {  
  Channel.fromPath("*.fastq.gz") | fastqc  
}
```

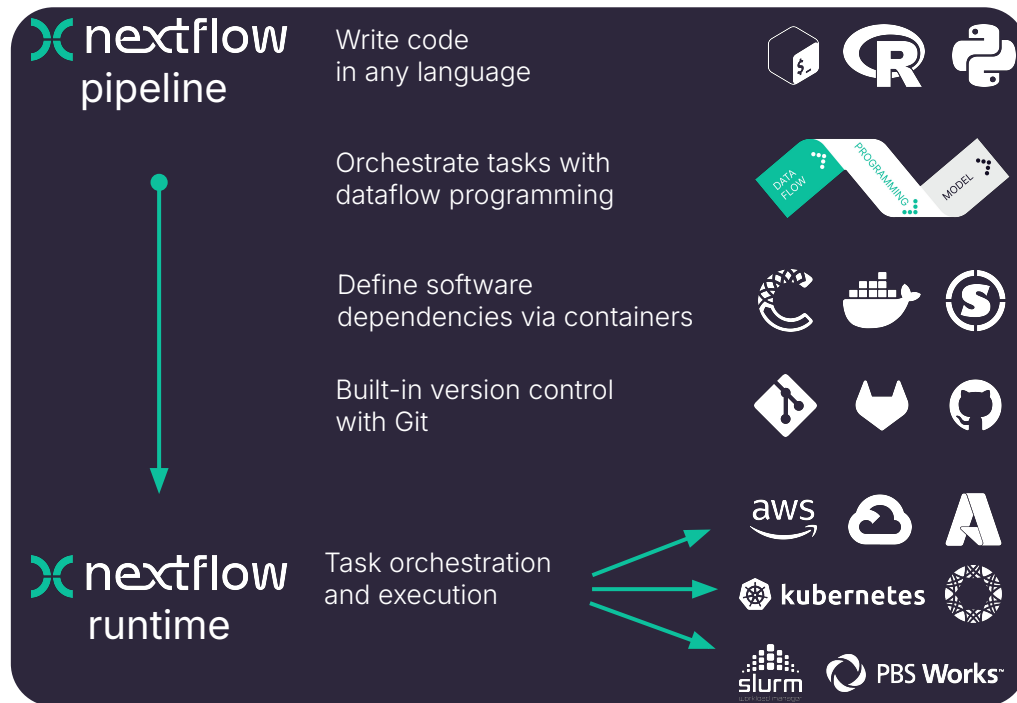
Nextflow

A reactive workflow framework and a programming DSL



Nextflow

Nextflow is a language, a runtime, and a community



Nextflow

Nextflow is a language, a runtime, and a community



Reproducible

Integration with code management tools, with versioned releases.



Portable

Docker, Singularity, Conda, works with most compute environments.



Scalable

5 samples on your laptop, 5k on an HPC or 5 million in the cloud.

(Opinionated) rule two
of bioinformatics

**Don't reinvent
the wheel...***

* please!

Before we start

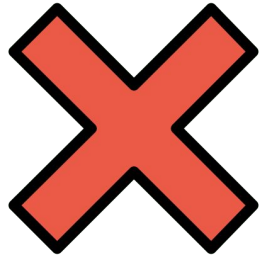
Some basic bioinformatics practise...

(Opinionated) rule
three of bioinformatics

**Never install
into your base
environment!**

Never install into your base

⚠ DO NOT RUN



```
conda install nextflow
```

Always install into a dedicated env

⚠ DO NOT RUN



```
conda create -n nextflow -c bioconda nextflow
```

Always install into a dedicated env

The purpose of *dedicated environments* is to ensure you don't get conflicting dependencies.

Tool 1: needs package v1.2

Tool 2: needs package v3.4

Install: Tool 2, replaces package v1.2, causes Tool 1 to stop working 😭!

Requirements for Nextflow

Requirements for Nextflow

To run any Nextflow/nf-core pipeline, you need only **three** things!

- ☕ Java
-  Nextflow
- A software environment tool ( onda, docker, singularity/apptainer etc.)



To get java and nextflow, create env!

```
conda create -n nf-core-dsl1 -c bioconda nextflow=22.10.6
```

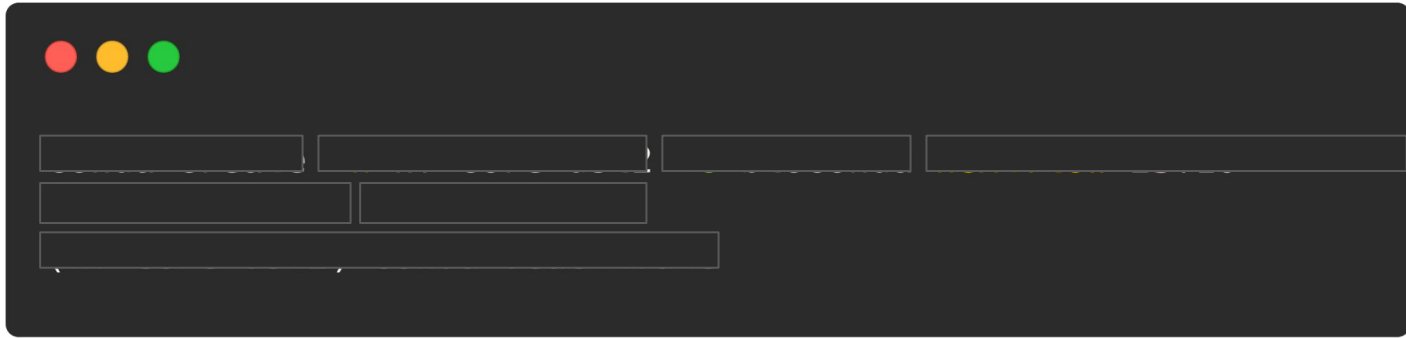
Name of environment

Specify to take software from a
bioconda **channel**

Software (and versions) to install into the dedicated environment

Lets practise!

Make another conda environment, but with the latest version of Nextflow. Activate it and deactivate it. (Hint: for version check GitHub)



Before we start!

Always practise data hygiene! Never just run in your home directory!



```
mkdir $HOME/acad-workshop/  
cd $HOME/acad-workshop/
```

More Info & Acknowledgements



<https://nf-co.re>



github.com/nf-core



nfcore.slack.com



youtube.com/nf-core



@nf_core@mstdn.science



[@nf_core](https://twitter.com/nf_core)

Some emojis designed by [OpenMoji](#) – the open-source emoji and icon project. License: [CC BY-SA 4.0](#)

Financial &
Infrastructure
Support:



HackMD

Speaker's
Support



[nf-core/core](#)
& community

Microbiome Sciences Group @
MAX-PLANCK-INSTITUT
FÜR EVOLUTIONÄRE ANTHROPOLOGIE



WSS
WERNER SIEMENS-STIFTUNG

